Department of Computer Science & Engineering
Microprocessor & Computer Architecture
UE24CS251B

MPCA Mini-project

Title: Smart Washroom System

Team Members:

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Abstract/Brief description of the project:

We developed an automated monitoring system for public restrooms using an Arduino Uno. The project focuses on improving hygiene and safety through a sensor-driven environment.

Hardware Integration & Connections: To support the high number of components, we established a robust power foundation using a **Breadboard**. We connected the **Arduino 5V and GND pins** to the breadboard's side rails and used **bridge wires** to link the top and bottom rails, ensuring a stable, shared power supply for every module. The wiring is organized so that the Arduino acts as the central brain, receiving analog and digital signals from the sensors and sending commands to the output devices (Buzzer and LEDs).

What we implemented:

- **Touchless Tap:** Uses an **IR Obstacle Sensor** to detect hands and simulate a hygienic, water-saving faucet.
- **Leakage Detection:** Uses a **Water Leakage Sensor (Probe + Module)** to monitor the floor for flooding or overflows, triggering a dedicated **LED**.
- **Odor & Safety Control:** Uses an **MQ-2 Gas Sensor** to track air quality, which activates a **Buzzer** if hazardous smells or gases are detected.
- **Smart Lighting:** Uses an **HC-SR04 Ultrasonic Sensor** to detect if a person is within range, automatically turning a second **LED** on for occupancy.

Priority-Based LCD Logic: The most important feature of the code is the **LCD Priority System**. Instead of showing all data at once, the screen follows a hierarchy to ensure the most critical information is seen first:

1. **Leakage Detected** (Highest Priority - Safety Alert)
2. **Bad Odor Alert** (Warning Alert)
3. **Tap Status: ON** (Activity Update)
4. **Motion Detected** (Occupancy Update)
5. **Monitoring...** (Idle Status)

By using this priority structure, the system ensures that maintenance issues like leaks are never ignored, even if other sensors are being triggered simultaneously.

Components used:

Arduino Uno R3, 16x2 LCD Display (with I2C Backpack), HC-SR04 Ultrasonic Sensor, MQ-2/7 Gas Sensor, Water Leakage Sensor (Probe + Module), FC-51 IR Obstacle Sensor, Active Buzzer, LEDs (x2), Resistors (x2), Breadboard, Jumper Wires and an USB Cable

Circuit Diagram:

